Week 01 Assignment

Data Quality, Evaluation, Scaling, and Encoding

Student name:

This is a small assignment that connects topics from Module 1, 2, and 3.
You must complete it in this Colab notebook.

You will need to use concepts that appeared in the videos:

- Module 1 and 2: basic descriptive statistics, proportions, confusion matrix, accuracy, precision, recall
- Module 3: standardization, min max scaling, nominal vs ordinal, one hot encoding, ordinal encoding, Euclidean and Manhattan distance

Please do not use any extra libraries beyond `pandas`, `numpy`.

---

0. Setup and Dataset

We will use a dataset that should have columns given below:

- `user_id`
- `age`
- `monthly_income` (numeric)
- `daily_screen_time_min` (numeric)
- `daily_app_opens` (numeric)
- `true_label` and `pred_label` for a binary classification task (0 or 1)
- `satisfaction_level` (for example: `Low`, `Medium`, `High`)
- `city_type` (for example: `Urban`, `Suburban`, `Rural`)

```
# Cell 1: Imports
import pandas as pd
import numpy as np
```

```
# Cell 2: Load the dataset (Already done for you)
df = pd.read_csv("https://drive.google.com/uc?export=download&id=1OmDDCh4MD1TtvAemnwVDyz5zwCIXJ220")

# Show first few rows
df.head()
```

	user_id	age	monthly_income	daily_screen_time_min	daily_app_opens	true_label	pred_label	satisfaction_level	city_type
0	1	43	3734.19	109	48	0	0	Medium	Suburban
1	2	49	2594.19	194	7	0	0	Low	Urban
2	3	19	3550.47	146	36	1	0	High	Rural
3	4	19	3821.18	287	14	1	0	High	Suburban
4	5	63	1750.84	66	46	0	0	Medium	Suburban

0.1 Check your dataset

1. Confirm that the dataset loaded correctly.
2. Check that you have at least these columns:
   - numeric: `age`, `monthly_income`, `daily_screen_time_min`, `daily_app_opens`
   - labels: `true_label`, `pred_label`
   - categorical: `satisfaction_level`, `city_type`

Part A - Module 1 and 2 Review

In this part you will do simple descriptive statistics and basic classification evaluation.

Q1. Descriptive statistics on a numeric feature

Choose one numeric column, for example `daily_screen_time_min`.

```
# Q1.1: Choose your numeric column here [We already write this ans]
num_col = "daily_screen_time_min"

df[num_col].describe()
```

> **Q1.2 Short answer: [Marks: 05]**
> Look at the count, mean, min, max, and standard deviation for your chosen column.
> In 2 to 3 sentences, comment on what you see.
> For example, does the max look very far from the mean, or does it look quite close?

Write your answer here:

Q2. Proportion of positive class

Use the `true_label` column, where 1 means "positive" and 0 means "negative".

```
# Q2.1: Compute proportion of positive class [We already write this ans]
label_col = "true_label"

positive_count = (df[label_col] == 1).sum()
total_count = df.shape[0]
positive_proportion = positive_count / total_count

print("Positive count:", positive_count)
print("Total samples:", total_count)
print("Proportion of positive class:", positive_proportion)
```

> **Q2.2 Short answer: [5 marks]**
> In 1 to 2 sentences, explain what this proportion tells you about your dataset.
> For example, is the dataset balanced between 0 and 1, or is one class much more common?

Write your answer here:

Q3. Confusion matrix and basic metrics

For this question, use:

- `true_label` as the actual label
- `pred_label` as the model prediction

```
# Q3.1: Manually compute TP, TN, FP, FN [We already write this ans]
true_col = "true_label"
pred_col = "pred_label"

tp = ((df[true_col] == 1) & (df[pred_col] == 1)).sum()
tn = ((df[true_col] == 0) & (df[pred_col] == 0)).sum()
fp = ((df[true_col] == 0) & (df[pred_col] == 1)).sum()
fn = ((df[true_col] == 1) & (df[pred_col] == 0)).sum()

print("TP:", tp)
print("TN:", tn)
print("FP:", fp)
print("FN:", fn)
```

```
# Q3.2: Compute accuracy, precision, recall [We already write this ans]
accuracy = (tp + tn) / (tp + tn + fp + fn) if (tp + tn + fp + fn) > 0 else 0

precision = tp / (tp + fp) if (tp + fp) > 0 else 0
recall = tp / (tp + fn) if (tp + fn) > 0 else 0

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
```

> **Q3.3 Short answer: [10 marks]**
> In 3 to 4 sentences, briefly comment on the model using these three metrics.
> For example, is the model catching most positives (high recall) or being careful when it predicts positive (high precision)?

Write your answer here:

Part B - Module 3: Scaling and Encoding

Now we will pick a few features and apply scaling and encoding.

Q4. Standardization and Min max scaling

Use one numeric column, `monthly_income`.

```
# Q4.1: Choose the numeric column [2 marks]
```

```
# Q4.2: Standardization with z-score [10 marks]
```

```
# Q4.3: Min max scaling implementation [10 marks]
```

> **Q4.4 Short answer: [3 marks]**
> Compare the standardized and min max scaled columns in 2 to 3 sentences.
> Mention what kind of range each one uses and how the numbers look.

Write your answer here:

Q5. One hot and ordinal encoding

We will use:

- `city_type` as a nominal feature
- `satisfaction_level` as an ordinal feature with order `Low` < `Medium` < `High`

```
# Q5.1: One hot encoding for city_type using pandas [10 marks]
```

```
# Q5.2: Attach one hot encoded columns to df [5 marks]
```

```
# Q5.3: Ordinal encoding for satisfaction_level [10 marks]
```

> **Q5.4 Short answer: [5 marks]**
> In 2 to 3 sentences, explain why one hot encoding is suitable for `city_type`
> and why ordinal encoding is suitable for `satisfaction_level`.

Write your answer here:

Part C - Module 3: Distances between users

For this small part we will work with vectors based on scaled numeric features.

Q6. Euclidean and Manhattan distance

Build 2D vectors for user 0 and user 1 using:

- `income_std`
- `daily_app_opens` (or its min max scaled version if you prefer)

```
# Q6.1: Build 2D vectors for first two users [We already write this ans]
vec_cols = ["monthly_income", "daily_app_opens"]

v1 = df.loc[0, vec_cols].values
v2 = df.loc[1, vec_cols].values

print("v1:", v1)
print("v2:", v2)
```

```
# Q6.2: Euclidean distance computation [5 marks]
```

```
# Q6.3: Manhattan distance computation [5 marks]
```

> **Q6.4 Short answer: [5 marks]**
> Which one is larger in your result, Euclidean or Manhattan distance
> and why does that usually happen based on their formulas?

Write your answer here:

Final Reflection [10 marks]

> In 4 to 6 sentences, describe how the three modules connect in this assignment.
> Mention:
>
> - One idea from Module 1 or 2 that you used
> - One idea from Module 3 that you used
> - How these ideas together help you understand a dataset more deeply

Write your reflection here:

End of Assignment

Before submitting:

- Run all cells from top to bottom.
- Check that all answer sections are filled.
- Instruction video অনুযায়ী আমাদের দেয়া Colab ফাইলটি থেকে প্রথম একটি Save copy in drive করে নিবা। এরপর Google colab এর মধ্যে কোডগুলো করবে এবং সেই ফাইলটি 'Anyone with the link' & 'View' Access দিয়ে ফাইলটির Shareble Link টি সাবমিট করবে।